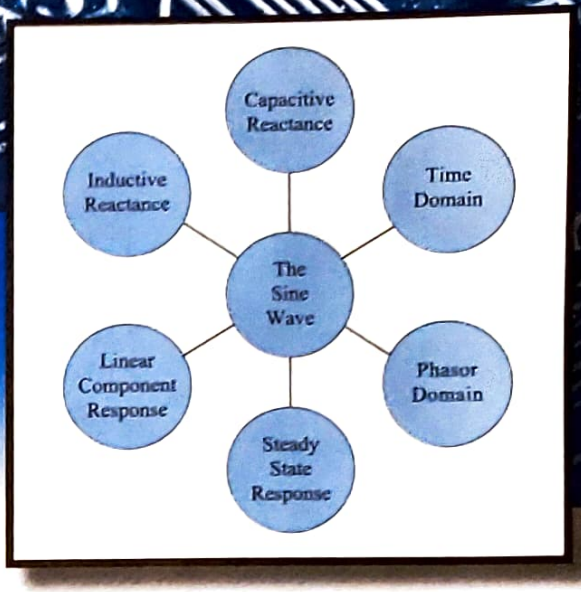
THE SINE WAVE AND LINEAR RESPONSE



"ANYTHING ONE PERSON CAN IMAGINE, ANOTHER PERSON CAN MAKE REAL."

—Jules Verne

Book:
Circuit Analysis Essentials:
A Signal Processing Approach by Bryant

CHAPTER OUTLINE

4.1 The Sine Wave as a Function of Time
4.2 Sine Wave Parameters
4.3 Instantaneous Value
4.4 Adding Sine Waves in the Time Domain
4.5 The Sine Wave in the Phasor Domain
4.6 Phasor Mathematics
4.7 The Phasor in Rectangular Form
4.8 Plotting Sine Waves with MATLAB
4.9 The Resistor in the Time Domain
4.10 The Resistor in the Phasor Domain
4.11 The Inductor in the Time Domain
4.12 The Inductor in the Phasor Domain
4.13 The Capacitor in the Time Domain
4.14 The Capacitor in the Phasor Domain

ENME 561-661 Note: There will not be a direct question from this file, but it is expected that you are familiar with this subject.

LEARNING OBJECTIVES

Upon successful completion of this chapter you will be able to:

- Express the sine wave as a function of time in the time domain.
- Determine the parameters of a sine wave.
- Calculate the instantaneous value of a sine wave.
- Express a sine wave as a phasor in the phasor domain.
- Convert a sine wave from the time domain to the phasor domain.
- Use an engineering calculator to perform mathematical operations in the phasor domain.

CHAPTER FOUR

- Express a phasor in polar or rectangular form.
- Calculate current and voltage for a resistor in the time domain.
- Calculate current and voltage for a resistor in the phasor domain.
- Calculate current and voltage for an inductor in the time domain.
- Calculate the reactance of an inductor at a give frequency.
- Calculate the reactance of a capacitor at a given frequency.
- Calculate current and voltage for an inductor in the phasor domain.
- Calculate current and voltage for a capacitor in the time domain.
- Calculate current and voltage for a capacitor in the phasor domain.
- Sketch a phasor diagram.

INTRODUCTION

The sinusoidal waveform is the most important electronic signal to be considered in this text, and the early sections of this chapter are devoted to the study of the sine wave. The sine wave is represented mathematically as a function of time in the time domain and as a phasor in the phasor domain. Sinusoidal voltage and current are typically expressed in the phasor domain to simplify mathematical calculations.

This chapter also introduces the linear components that are utilized by engineers and engineering technologists to design the electronic circuits that create, process, and transmit electronic signals. A linear component is one for which the current is always proportional to the voltage across the device. The resistor, inductor, and capacitor are linear components. This chapter will consider the physical nature of each component, derive its mathematical representation or mathematical model, and determine how each of the components responds to sinusoidal waveforms. Future chapters will demonstrate how these components are interconnected to create electronic circuits and how such circuits may be analyzed and designed.

4.1 THE SINE WAVE AS A FUNCTION OF TIME

The sine wave is the most important electronic signal waveform to be considered in the analysis and design of analog electronic circuits. The sine function is often presented in mathematics texts as a function of an independent variable often designated by the letter x.

The **amplitude** of a sine wave is the value of the function at the point of maximum displacement from the zero axis.

The sine function is typically expressed in mathematics texts as:

$$f(x) = A \sin(x) \tag{4.1}$$

where A is the amplitude of the waveform and x is the independent variable.

A MATLAB plot of $f(x) = A \sin(x)$ is shown in Figure 4–1.

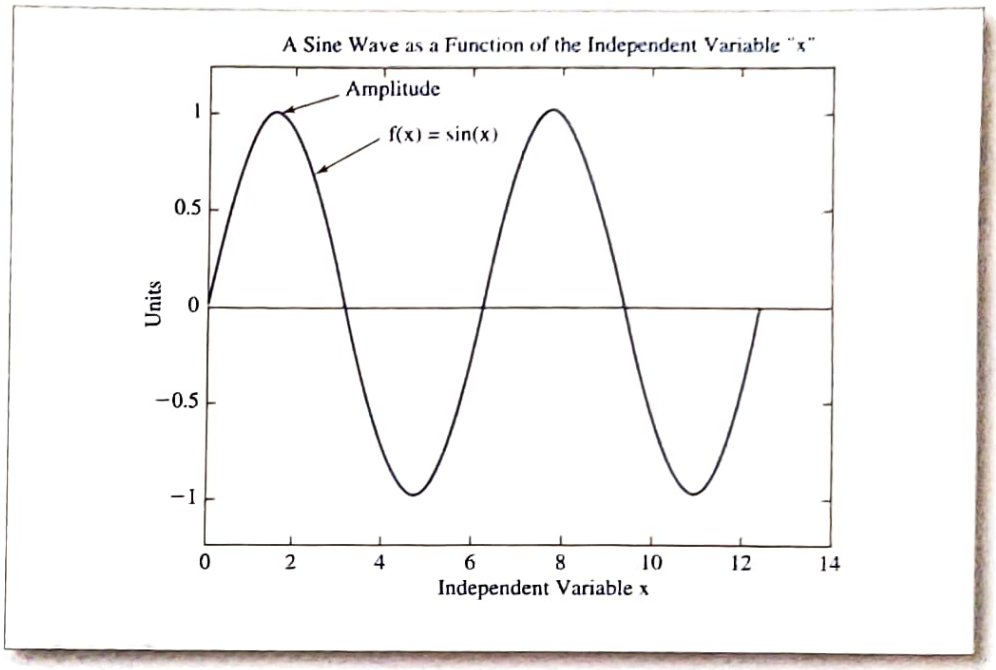


FIGURE **4–1**

A Sine Wave as a Function of the Independent Variable "x"

A waveform that is expressed as a function of time is said to be expressed in the **time domain.**

Figure 4–2 shows a sinusoidal voltage signal as a function of time represented in the time domain.

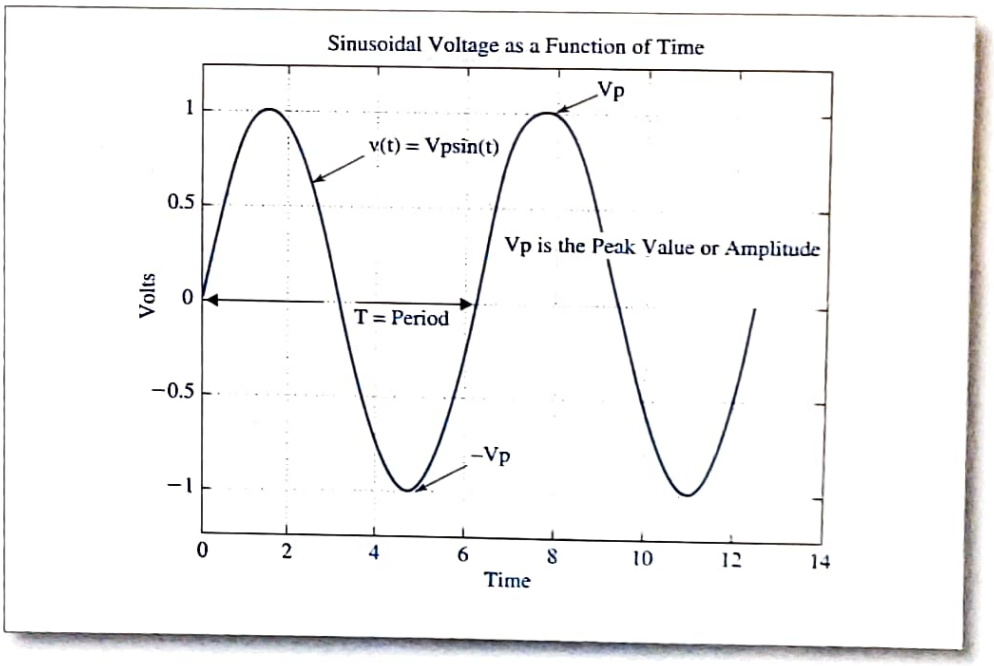


FIGURE **4–2**

A Sinusoidal Voltage as a Function of Time

4.2 SINE WAVE PARAMETERS

Several important observations can be made regarding the sine wave in Figure 4–2 that are common to all sine waves. The sine wave repeats itself at regular intervals and is

therefore said to be a **periodic waveform.** Each complete repetition is called a **cycle** of the waveform. The time required to complete one cycle of the waveform is called the **period** and is represented by the symbol T.

The **cyclic frequency** of a periodic waveform is represented by the symbol f and is defined as the number of cycles that occur in one second of time. Frequency (f) is measured in units of hertz (Hz) to honor the German physicist Heinrich Hertz (1857–1894).

The cyclic frequency of a sine wave is the reciprocal of the period such that:

$$f = \frac{1}{T} \tag{4.2}$$

Maximum displacement from the zero axis may be referred to as amplitude; however, electronic engineers typically refer to the maximum displacement as the **peak value** of the waveform. The terms *amplitude* and *peak value* describe the same quantity so that the terms may be used interchangeably.

Designating the peak value of a sinusoidal voltage waveform as V_P and expressing the voltage as a function of time, we obtain:

$$v(t) = V_P \sin(t) \tag{4.3}$$

Although the expression $v(t) = V_P \sin(t)$ in Equation (4.3) indicates that the wave shape is sinusoidal and indicates the peak value or amplitude, other important features or parameters of electronic signals are not included in this expression.

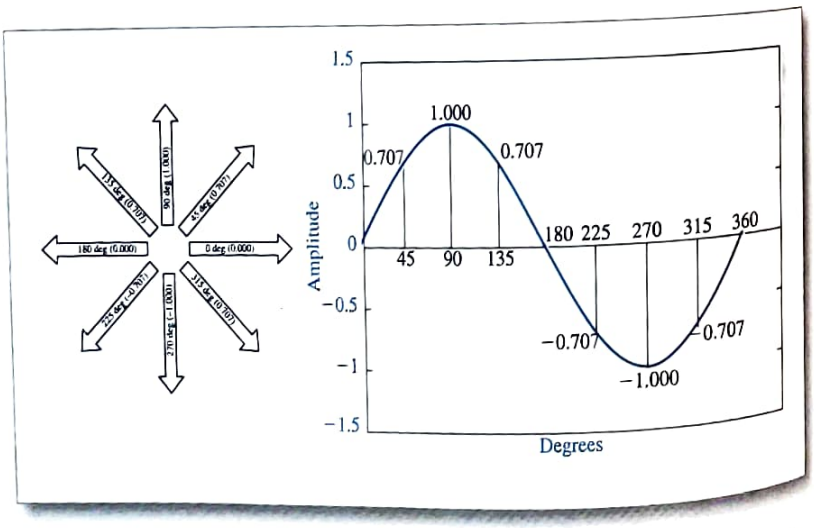
The frequency of an electronic signal is an important consideration in the analysis and design of analog electronic circuits. The expression in Equation (4.3) must be modified to indicate the frequency of the sine wave.

A **vector** is a mathematical term used to indicate quantities that are defined by both magnitude and direction or position.

The magnitude of a vector is proportional to its length, and its position is typically specified by an angular displacement from a given reference point. If a unit vector (magnitude = 1) is revolved counterclockwise about a fixed point, the projection of the vector (distance from the tip of the vector to the reference axis) describes the sine function as illustrated in Figure 4–3.

FIGURE **4–3**

A Revolving Unit Vector Generating a Sine Wave



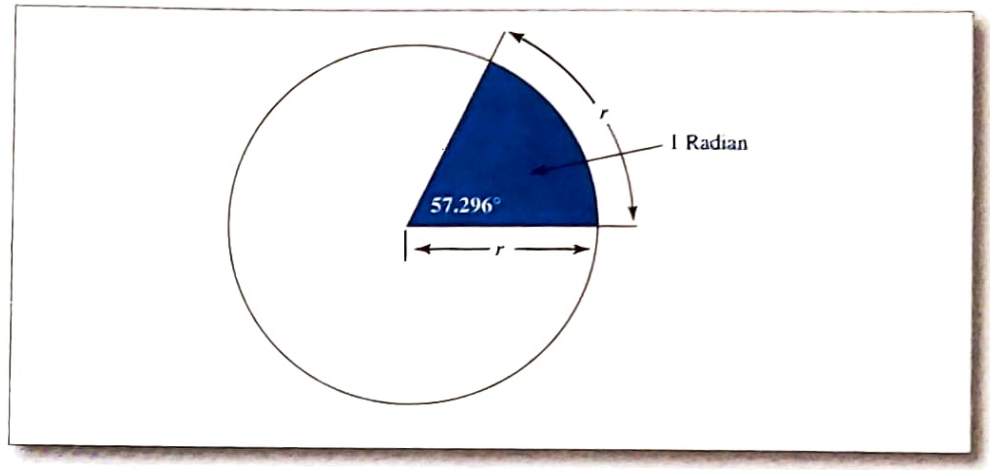
Notice that the projection of the unit vector through 360° of revolution generates one complete cycle of the sine wave, the point at which the waveform begins a new

repetition. Although angular position may be measured in degrees, the radian provides an alternative unit of measure that is often preferred by engineers and scientists.

The *radian* is derived from the geometric relationship between the circumference and the radius of a circle. It is defined as the angle subtended by an arc equal to the length of the radius along the circumference, and that angle, regardless of the size of the circle, is 57.296° as indicated in Figure 4–4.

FIGURE 4–4

One Radian is Equivalent to 57.296 Degrees



Since the circumference of a circle is 2π times the radius, it follows that 2π radians is equivalent to 360°.

$$2\pi \text{ radians} = 360° \tag{4.4}$$

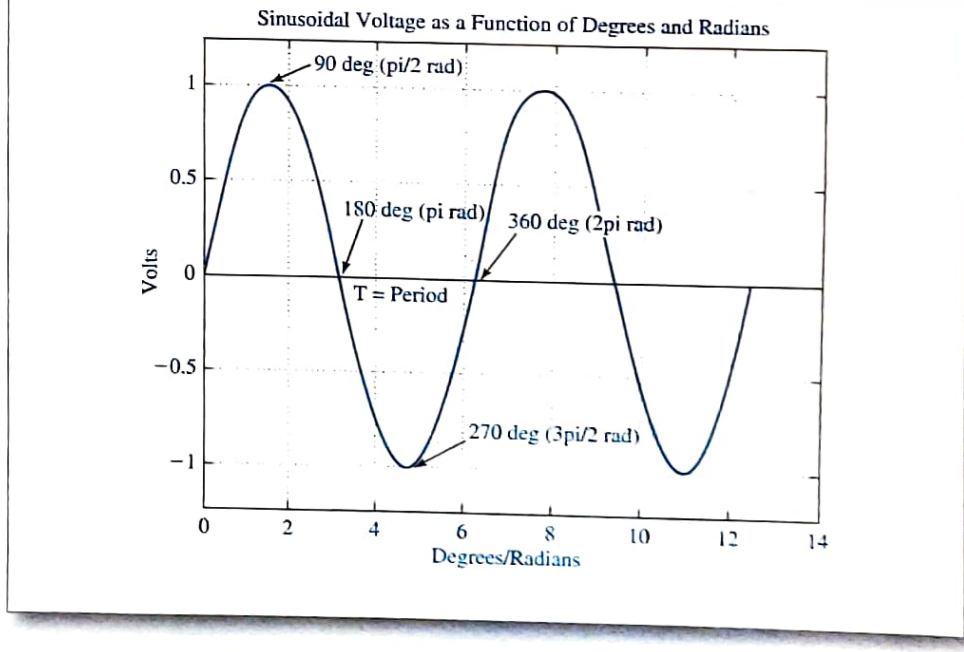
Angular displacement, therefore, may be indicated in degrees or radians. The radian measure is traditionally chosen to represent the angular displacement of the unit vector as it revolves to generate the sine wave. Therefore, a unit vector that revolves through 2π radians will describe one complete cycle of the sine wave.

The sine wave shown in Figure 4–5 is shown as a function of angular displacement measured in both degrees and radians.

MATLAB

FIGURE 4–5

A Sinusoidal Voltage as a Function of Degrees and Radians



The equivalent values of degrees and radians at the points of maximum displacement and at the zero crossing points are shown below:

1. 2π radians = 360° ($V = 0$)
2. π radians = 180° ($V = 0$)
3. $\frac{\pi}{2}$ radians = 90° ($V = V_P$)
4. $\frac{3\pi}{2}$ radians = 270° ($V = -V_P$)

Our objective, however, is to represent the sinusoidal waveform as a function of time, not as a function of degrees or radians. We therefore need to devise a procedure that will indicate the angular position of the revolving vector illustrated in Figure 4–3 as a function of time.

Consider that if the angular velocity of the revolving vector in Figure 4–3 is known and the time that the vector has been revolving is also known, the angular position of the vector can be determined. This is analogous to the way that the distance traveled may be determined if the speed or velocity and the time of travel are known (distance = velocity × time).

The angular velocity of a revolving vector whose projection generates a sine wave is given the name omega and is represented by the symbol ω. Although omega (ω) is technically a measure of angular velocity having the units radians/second (rad/s), it is also referred to as angular frequency or radian frequency. The quantity omega will be referred to as **radian frequency** throughout this text.

The angular position of the revolving vector can be determined by multiplying the **radian frequency** (ω) of the revolving vector by **elapsed time** (t). The sinusoidal voltage waveform may therefore be expressed as a function of time (t) and omega (ω) to obtain:

$$v(t) = V_p \sin(\omega t) \tag{4.5}$$

Caution must be exercised to avoid confusing radian frequency (ω), which is measured in units of rad/s, with cyclic frequency (f), which is measured in units of hertz (Hz).

As noted, the revolving vector will travel 2π radians during each complete revolution, so that angular velocity or radian frequency may be defined as:

$$\omega = \frac{2\pi}{T} \text{ (rad/s)} \tag{4.6}$$

An equivalent expression may be derived by substituting $f = \frac{1}{T}$ in Equation (4.6) to obtain:

$$\omega = 2\pi f \tag{4.7}$$

The sinusoidal voltage signal as a function of time and cyclic frequency may be expressed as:

$$v(t) = V_p \sin(2\pi ft) \tag{4.8}$$

where f is the cyclic frequency measured in Hz.

It is important to note that Equations (4.5) and (4.8) are expressions that indicate the peak value (amplitude) as well as the frequency of the waveform. Although the expressions

in Equations (4.5) and (4.8) are equivalent, Equation (4.5) is the expression preferred by engineers and will be used most often throughout this text.

Figure 4–6 illustrates the relationship between omega, period, and cyclic frequency (Hz).

MATLAB

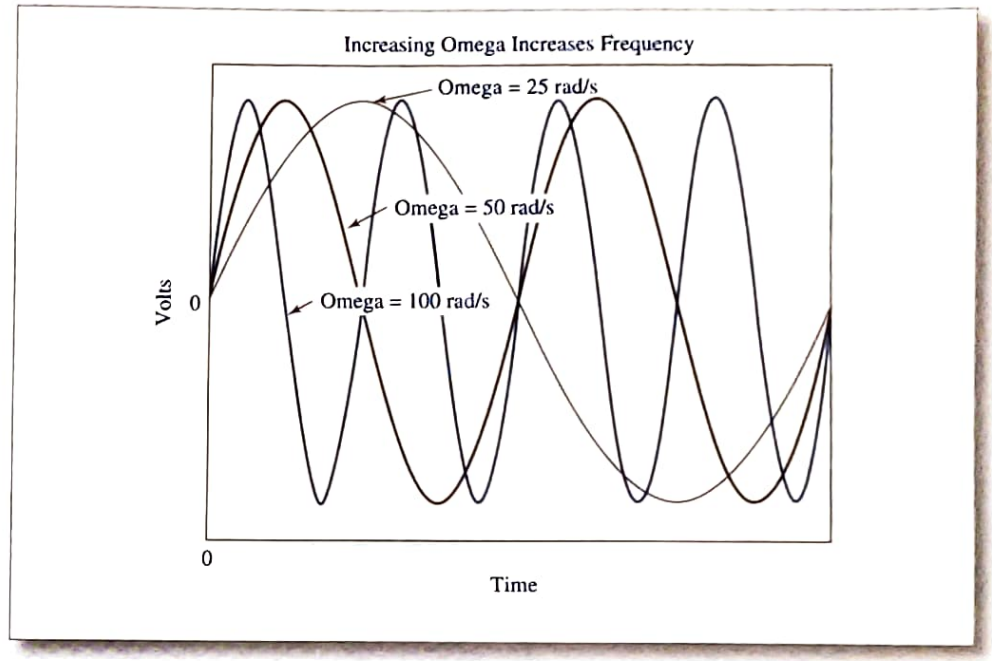


FIGURE 4–6

Increasing Omega Increases Frequency

Observe in Figure 4–6 that increasing the radian frequency (ω) decreases the period (T) and increases the cyclic frequency (f).

Many circuits that are used to process electronic signals may cause the signals to be delayed or advanced in time. The analytical expression for a sinusoidal signal waveform should therefore include information regarding any time shifts that may occur. The shift may also be indicated in angular measurement of degrees, rather than a time shift measured in seconds, which is simply a matter of preference.

Phase angle is defined as the angular displacement of a waveform with respect to a specified reference. The phase angle of electronic signals is measured in degrees and designated by the symbol θ (theta). A positive value for θ has the effect of advancing the signal in time, and a negative value of θ delays the signal in time.

The phase angle of a waveform has meaning only if a reference is specified. The reference may be simply specified to be 0°. The phase angle may also be referenced to another sine wave that exists in the circuit.

Phase difference is a measure of the angular displacement of a wave from the specified reference wave. Phase difference is determined by subtracting the phase angle of a given waveform from the phase angle of the reference waveform. The phase angle of the reference is typically considered to be 0°.

The phase difference between the two sine waves shown in Figure 4–7 is 90° as indicated.

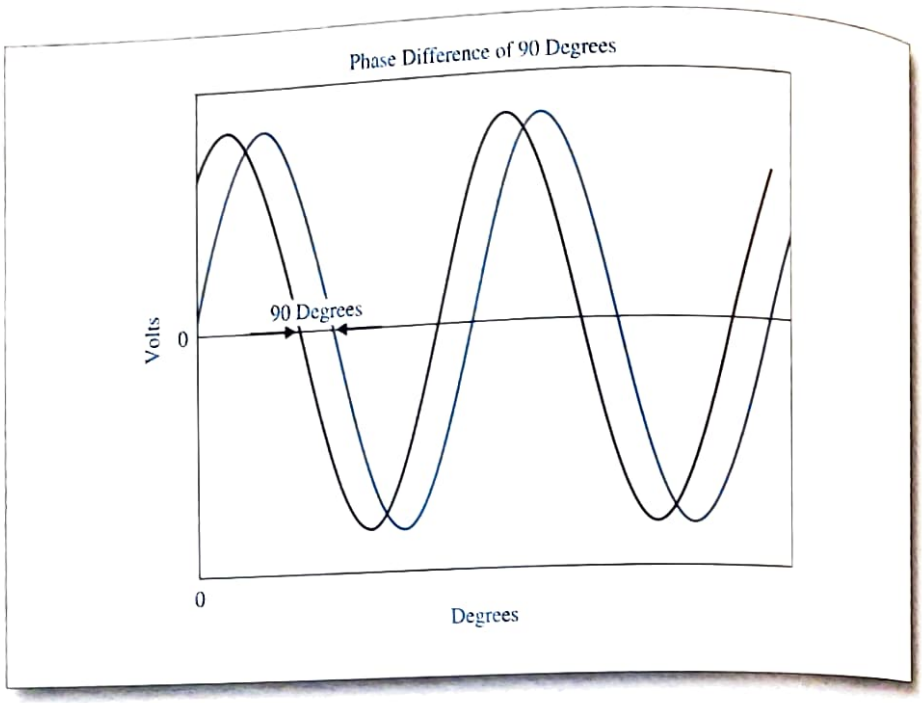


FIGURE 4–7

Waveforms Whose Phase Difference is 90 Degrees

Note that the sine waves in Figure 4–7 do not cross the zero axis at the same point. The difference between the crossover points measured in degrees is 90° as indicated.

The general expression for a sinusoidal voltage waveform as a function of time that includes phase angle may be expressed as:

$$v(t) = V_p \sin(\omega t \pm \theta) \tag{4.9}$$

The expression in Equation (4.9) represents the analytical expression of a sinusoidal voltage signal from which the peak value, period, frequency, and phase angle may be determined.

Quantities that vary over time are, by convention, represented by lowercase symbols, and quantities that are constant or do not vary over time are represented by uppercase symbols. This convention will be followed throughout the text. Therefore, v is understood to be equivalent to $v(t)$, and i is equivalent to $i(t)$. The expression in Equation (4.9) may therefore be written as:

$$v = V_p \sin(\omega t \pm \theta) \tag{4.10}$$

and a similar expression for sinusoidal current as a function of time may be written as:

$$i = I_p \sin(\omega t \pm \theta) \tag{4.11}$$

EXAMPLE 4–1 Expressing Voltage As a Function of Time

The peak value of a sinusoidal voltage waveform is given as 3 V. The phase angle of the waveform is 90°. Express the waveform as a function of time.

✓SOLUTION

The peak value of the waveform is given as 3 V and the phase angle is given as 90°. Substituting the given values into the general expression in Equation (4.10), the waveform may be expressed as:

$$v = 3 \sin(\omega t + 90°) \text{ V}$$

The waveform is shown in the MATLAB plot in Figure 4–8.

MATLAB

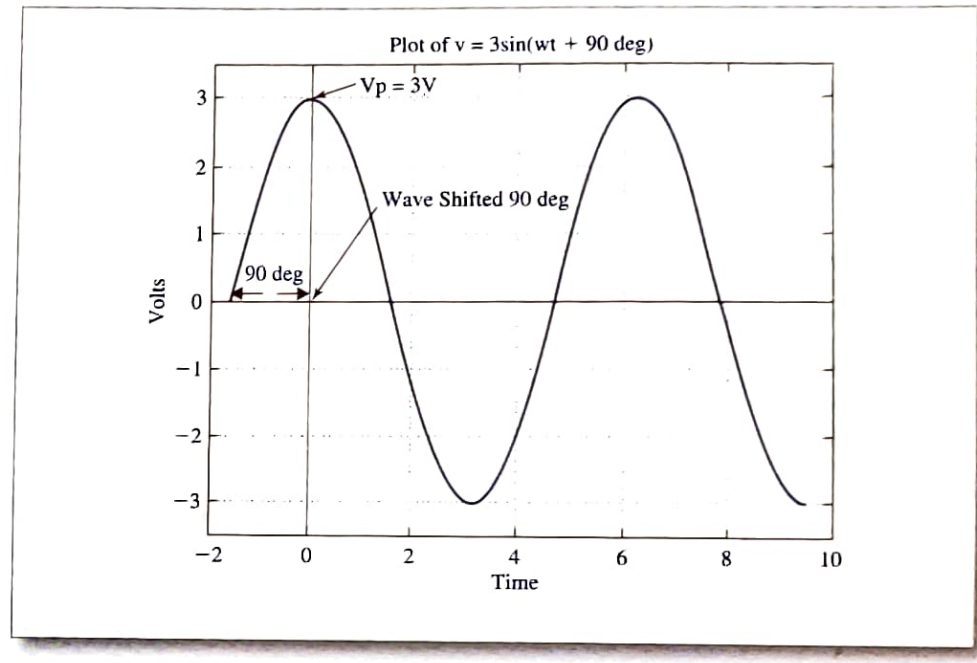


FIGURE **4–8**
Waveform for Example 4–1

Note that the waveform in Figure 4–8 does not begin a new cycle at the origin (specified reference) at time ($t = 0$). The waveform has been shifted to the left by 90° (advanced in time). The 90° phase shift is also indicated by the given analytical expression $v = 3 \sin(\omega t + 90°)$. As noted, the phase shift of a waveform is typically expressed in degrees rather than units of time.

EXAMPLE 4–2 Determining Phase Difference between Two Waveforms

Determine the phase difference between the given voltage waveforms, designating v_1 as the reference waveform. $v_1 = 1.5 \sin(\omega t + 90°)$ V; $v_2 = 6 \sin(\omega t + 60°)$ V.

✓SOLUTION

The phase difference is calculated as:

$$\theta = (\theta_1 - \theta_2) = (90° - 60°) = 30°$$

The two voltage waveforms are shown in Figure 4–9.

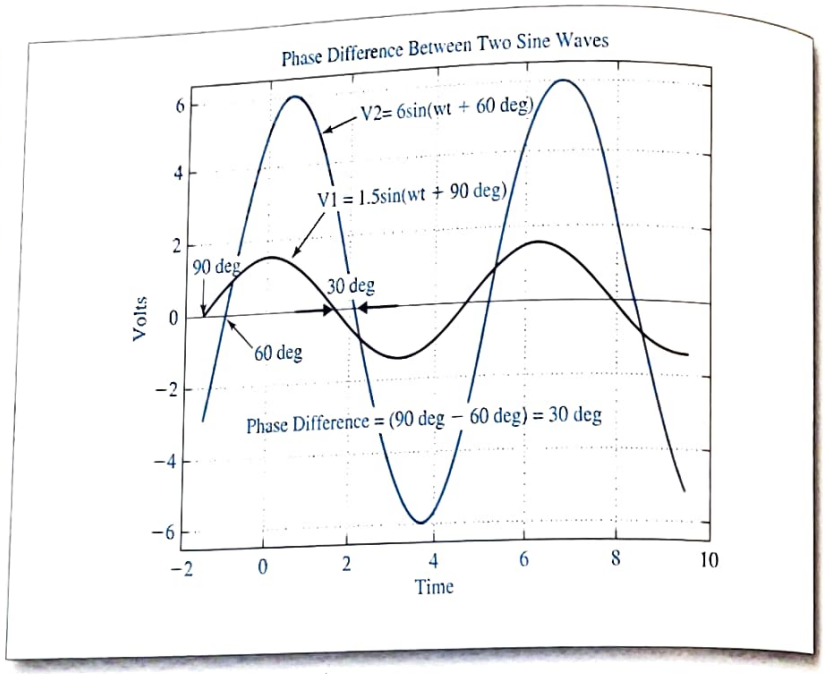


FIGURE 4–9
Waveforms for Example 4–2

Note that v_1 has been shifted 90° to the left of the origin ($t = 0$) and v_2 has been shifted 60° to the left of the origin. The phase difference between v_1 and v_2 is 30° as indicated in Figure 4–9. Note that the positive peak of v_1 occurs before the peak of v_2.

We therefore may say that v_1 *leads* v_2 by 30°. *Note that the amplitude of the waveforms is not considered when determining phase difference.*

EXAMPLE 4–3 Determining the Parameters of a Sinusoidal Voltage Signal

A sinusoidal voltage signal is expressed as $v = 12 \sin(1000t)$ V. Determine the value of peak voltage (V_P), amplitude, radian frequency (ω), frequency (f), period (T), and phase angle (θ).

✓SOLUTION

Comparing the given expression to the general expression of a sinusoidal voltage waveform expressed as:

$$v = V_p \sin(\omega t \pm \theta)$$

observe that the peak voltage or amplitude is 12 V. The radian frequency (ω) is 1000 rad/s, and the phase angle (θ) is 0°. The frequency (f) may be calculated as:

$$f = \frac{\omega}{2\pi} = \frac{1000}{2\pi} = 159.15 \text{ Hz}$$

In summary, the parameters of the voltage $v = 12 \sin(1000t)$ are:

$V_P = 12$ V; $\omega = 1000$ rad/s; $T = 6.283$ ms; $f = 159.15$ Hz; and $\theta = 0°$.

A MATLAB plot of the given function is shown in Figure 4–10.

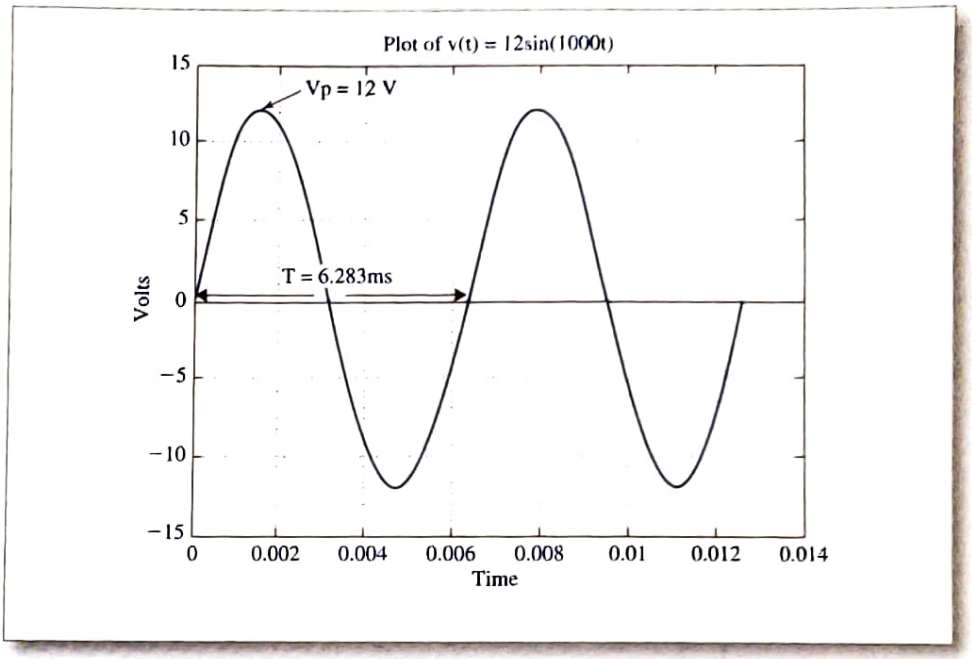


FIGURE 4–10
Waveforms for Example 4–3

PRACTICE EXERCISES 4–1

a. Given $v = 25 \sin(1000t)$ V, determine the indicated parameters: V_P, A (amplitude), ω, f, T, and θ.

b. Given $v = 16 \sin(3000t + 81°)$ V, determine the indicated parameters: V_P, A (amplitude), ω, f, T, and θ.

✓ANSWERS

a. $V_P = 25$ V, $A = 25$ V, $\omega = 1000$ rad/s, $f = 159.15$ Hz, $T = 6.28$ ms, $\theta = 0°$

b. $V_P = 16$ V, $A = 16$ V, $\omega = 3000$ rad/s, $f = 477.46$ Hz, $T = 2.09$ ms, $\theta = 81°$

(Complete solutions for the Practice Exercises can be found in Appendix A.)

4.3 INSTANTANEOUS VALUE

The sine wave is a continuous function of time and will have a unique value that can be determined for every instant of time.

The **instantaneous value** of a function of time is the unique value of the function at a specified time.

The instantaneous value of a sine wave can be either positive or negative, and the absolute value is less than or equal to the peak value of the waveform.

EXAMPLE 4–4 Calculating Instantaneous Values

Given $v = 12 \sin(1000t)$ V, calculate the instantaneous value of the voltage at $t =$ 1.57 ms, $t = 2$ ms, and $t = 4.7$ ms. Obtain a MATLAB plot to verify the solution.

✓SOLUTION

At $t = 1.57$ ms

Substituting the given value of time (1.57 ms) and evaluating the argument of the sine function, we obtain:

$$v = 12 \sin(1000 \times 1.57 \text{ ms}) = 12 \sin(1.57 \text{ rad})$$

With the calculator in *radian mode*, evaluate $\sin(1.57 \text{ rad}) = 0.999$ so that:

$$v = 12(0.99999) = 12 \text{ V (rounded)}$$

Note that at time $t = 1.57$ ms the function is at its *positive peak value* (12 V). This is true because 1.57 ms is one-fourth of the period (6.283 ms / 4 = 1.57 ms), which is the time at which the positive peak will occur. The argument of the sine function is 1.57 rad, which is equivalent to $\pi/2$ rad or 90°, which is the angle at which the positive peak value occurs.

At $t = 2$ ms

$$v = 12 \sin(1000 \times 2 \text{ ms}) = 12 \sin(2 \text{ rad}) = 12(0.9093) = 10.91 \text{ V}$$

At $t = 4.71$ ms

$$v = 12 \sin(1000 \times 4.71 \text{ ms}) = 12 \sin(4.71 \text{ rad}) = -12 \text{ V}$$

Note that at time $t = 4.71$ ms the function is at its *negative peak value* (−12 V). This is true because 4.71 ms is three-fourths of the period (0.75 T), which coincides with the negative peak value for the given sine wave. Note that 4.71 rad = $3\pi/2$ rad = 270°.

Figure 4–11 illustrates the instantaneous values calculated in Example 4–4.

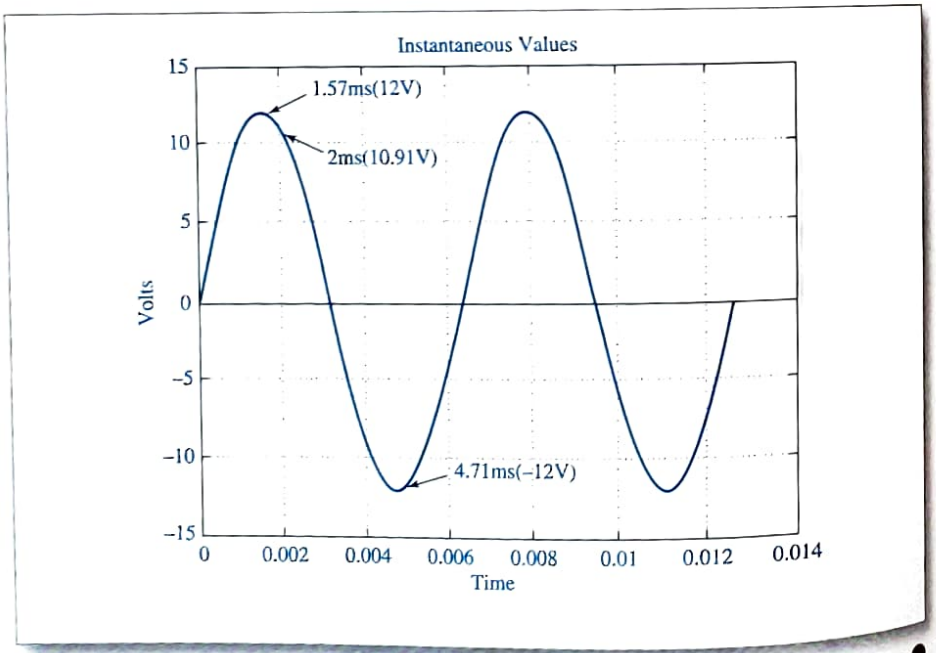


FIGURE 4–11
Instantaneous Values for Example 4–4

PRACTICE EXERCISES 4–2

Determine the instantaneous value of the given sinusoidal signals at the specified times.

a. $v = 6\sin(2500t)$ V at 1.3 ms

b. $i = 9\sin(6000t)$ mA at 250 μs

c. $v = 24\sin(4000t + 30°)$ V at 0.5 ms

✓ANSWERS

a. −0.65 V

b. 8.98 mA

c. 13.89 V



The calculator can be used to convert 30° to radians as required by Practice Exercise 4.2c by selecting RADIAN mode. Enter 30° into the display and press ENTER. The result indicates that 30° is equivalent to 523.6 × 10-3 rad. By selecting EXACT for the EXACT/APPROXIMATE mode, the calculator returns the exact answer $\pi/6$ rad.

(Complete solutions to the Practice Exercises can be found in Appendix A.)

Often engineers and scientists solve technical problems assisted by engineering calculators, as well as mathematically based computer software such as MATLAB and computer simulation programs such as MultiSIM. Calculators and computers are indispensable tools and are used extensively throughout this text. However, such tools are productive only when the user has a clear understanding and appreciation of the mathematics upon which the computations are based.

4.4 ADDING SINE WAVES IN THE TIME DOMAIN

During the analysis of electronic circuits it is necessary to perform mathematical operations involving sinusoidal voltages and currents. This section illustrates how the sum of two sine waves may be determined in the time domain using trigonometric identities.

EXAMPLE 4–5 Addition of Sine Waves Using Trigonometry

Determine the sum of the two sinusoidal voltages v_1 and v_2 using trigonometric identities.

$$v_1 = 1\sin(1000t + 60°); \quad v_2 = 1\sin(1000t + 30°)$$

✓SOLUTION

The trigonometric identity that expresses the sum of two sine functions is:

$$\sin(\alpha) + \sin(\beta) = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

This identity may be used to find the sum of the given sine waves as follows:

Let $\alpha = (1000t + 60°)$ and $\beta = (1000t + 30°)$, then:

$$\frac{(\alpha+\beta)}{2} = \frac{1}{2}[(1000t + 60°) + (1000t + 30°)] = 1000t + 45°$$

$$\frac{(\alpha-\beta)}{2} = \frac{1}{2}[(1000t + 60°) - (1000t + 30°)] = 15°$$

so that: $$2 \sin\frac{(\alpha+\beta)}{2}\cos\frac{(\alpha-\beta)}{2} = 2 \sin(1000t + 45°)\cos(15°)$$

Using the calculator, we find $\cos(15°) = 0.9659$

so that: $$2 \sin(1000t + 45°)\cos(15°) = 2 \sin(1000t + 45°)(0.9659)$$

and: $$v_1 + v_2 = 1.932 \sin(1000t + 45°) \text{ V}$$

Note that both of the given signals have the same peak value (1V). Adding sine waves that have different amplitudes using trigonometry greatly increases the complexity of the procedure.

4.5 THE SINE WAVE IN THE PHASOR DOMAIN

We have seen how sinusoidal expressions may be added in the time domain using trigonometric identities. Sine waves may also be added using a method that is based upon the use of *phasor mathematics* and an appropriate engineering calculator. Phasor mathematics, as we will see, is similar to the mathematics involving complex numbers.

Recall that the general expression of a sinusoidal voltage as a function of time is expressed as:

$$v = V_p \sin(\omega t \pm \theta)$$

Since we know that the waveform is sinusoidal, and if the frequency is also known, then the signal can be described using a shorthand mathematical expression:

$$\mathbf{V} = V_p \angle \theta_v \tag{4.12}$$

The quantity in Equation (4.12) is referred to as a *phasor in polar form*. Observe that the phasor representation of the sine wave specifies only the peak value (amplitude) and phase angle. Throughout this text, phasors will be represented by bold-faced uppercase symbols.

Although phasors, vectors, and complex numbers are similar entities, the term phasor is preferred when referring to electronic voltage or current signals.

A signal expressed using phasor notation is said to be expressed in the **phasor domain.**

When converting waveforms to their phasor domain expressions, it is important to keep in mind that the phasor domain is meaningful only for waveforms that are sinusoidal and that have the same frequency.

EXAMPLE 4–6 Converting Signals from Time Domain to Phasor Domain

Convert the given voltage waveform from the time domain to the phasor domain and sketch a plot of the phasor in polar form in the phasor domain.

$$v = 12 \sin(8000t + 38°) \text{ V (time domain)}$$

✓SOLUTION

The polar form in the phasor domain need indicate only the peak value and phase angle for the phasor. Therefore, the given sinusoidal voltage may be represented in the phasor domain as:

$$\mathbf{V} = 12 \angle 38° \text{ V (phasor domain)}$$

The plot of the phasor in polar form in the phasor domain is shown in Figure 4–12.

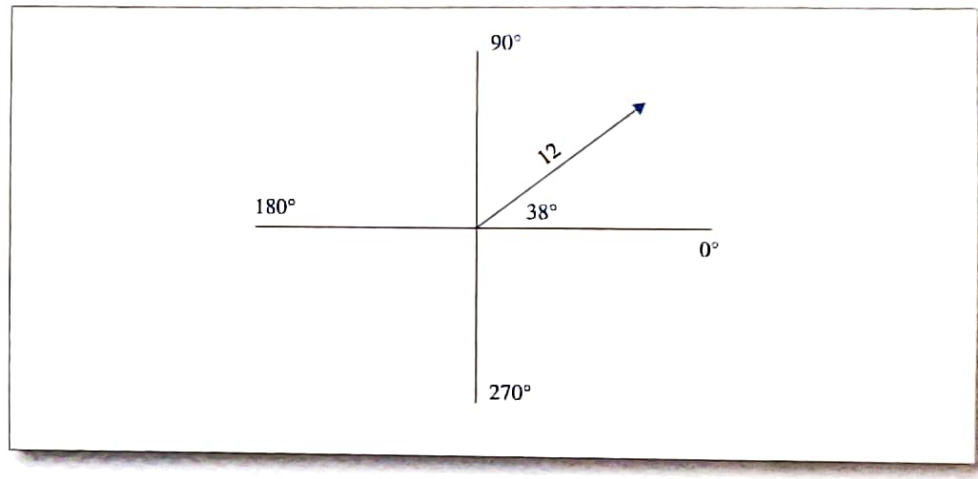


FIGURE 4–12
Phasor $\mathbf{V} = 12 \angle 38°$ V Plotted in Polar Form in the Phasor Domain

Note that the axes in the phasor domain diagram are labeled in degrees when expressing the phasors in polar form. Although the instantaneous values of a sinusoidal waveform cannot be determined from its phasor domain representation, this is a small price to pay for the advantages that are gained by performing mathematical computations in the phasor domain.

4.6 PHASOR MATHEMATICS

Mathematical operations in the phasor domain require that each voltage and current be represented in its phasor form.

An engineering calculator greatly simplifies calculations in the phasor domain, as illustrated in the following examples.

EXAMPLE 4–7 Adding Sine Waves in the Phasor Domain

Determine the sum of the sinusoidal voltages given in Example 4–5 using phasor mathematics. Express the final answer in the time domain. Sketch a phasor diagram of the two voltages and their sum.

$$v_1 = 1\sin(1000t + 60°)\text{ V};\quad v_2 = 1\sin(1000t + 30°)\text{ V}$$

✓SOLUTION

Each of the given sine waves must be converted from the time domain to the phasor domain as follows:

$$v_1 = 1\sin(1000t + 60°)\text{ V (time domain)} \rightarrow \mathbf{V}_1 = 1\angle 60°\text{ V (phasor domain)}$$

$$v_2 = 1\sin(1000t + 30°)\text{ V (time domain)} \rightarrow \mathbf{V}_2 = 1\angle 30°\text{ V (phasor domain)}$$



The following calculator keystrokes may be used for the TI-89 calculator.

Set the calculator COMPLEX FORMAT mode to POLAR and ANGLE mode to DEGREES. *Each phasor must be placed in parentheses in the calculator.* The calculator symbol for the angle of the phasor (∠) is the second function of the [EE] key. The symbol for degree (°) is *not required.*

The calculation is keyed as: ([1] ∠ [6][0]) [+] ([1] ∠ [3][0]) [ENTER] to obtain: 1.932 ∠ 45.

The phasor 1.932 ∠ 45 V can be converted to the time domain if desired by reinserting the information that was eliminated when the signals were converted to the phasor domain to obtain:

$$v_1 + v_2 = 1.932\sin(1000t + 45°)\text{ V}$$

Note that the sum of $v_1 + v_2$ obtained in Example 4–7 is the same result obtained in Example 4–5. The graphical representation of the given phasors and their sum is shown in the phasor diagram in Figure 4–13.

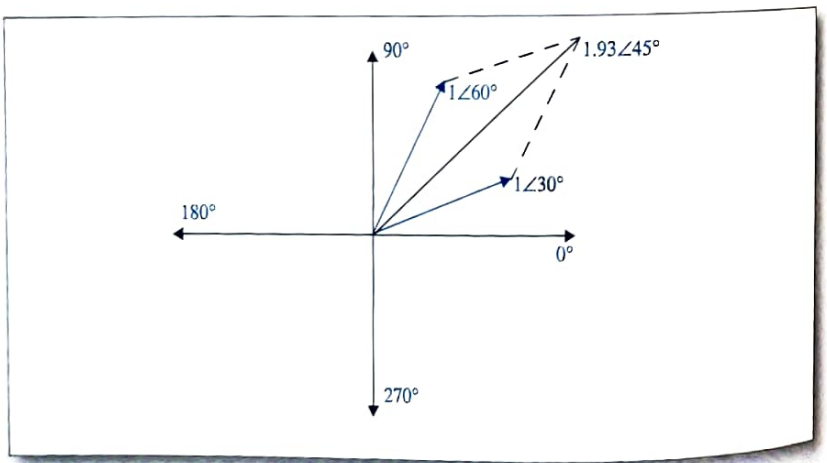


FIGURE 4–13
Phasor Diagram for Example 4–7

EXAMPLE 4–8 Adding Sine Waves in the Phasor Domain

Add the given sine waves using phasor mathematics. Express the final sum in polar form in the phasor domain and in the time domain. Sketch a phasor diagram of the two voltages and their sum.

$$v_1 = 18\sin(10t + 36°)\ \text{V};\quad v_2 = 7\sin(10t + 53°)\ \text{V}$$

✓SOLUTION

The given voltage signals may be converted to the phasor domain as:

$$v_1 = 18\sin(10t + 36°)\ \text{V (time domain)} \rightarrow \mathbf{V}_1 = 18\angle 36°\ \text{V (phasor domain)}$$

$$v_2 = 7\sin(10t + 53°)\ \text{V (time domain)} \rightarrow \mathbf{V}_2 = 7\angle 53°\ \text{V (phasor domain)}$$



The calculation is keyed as: (18 ∠ 36) + (7 ∠ 53) ENTER to obtain: 24.78 ∠ 40.74

so that: $\mathbf{V}_3 = \mathbf{V}_1 + \mathbf{V}_2 = 18\angle 36°\ \text{V} + 7\angle 53°\ \text{V} = 24.78\angle 40.74°\ \text{V}$

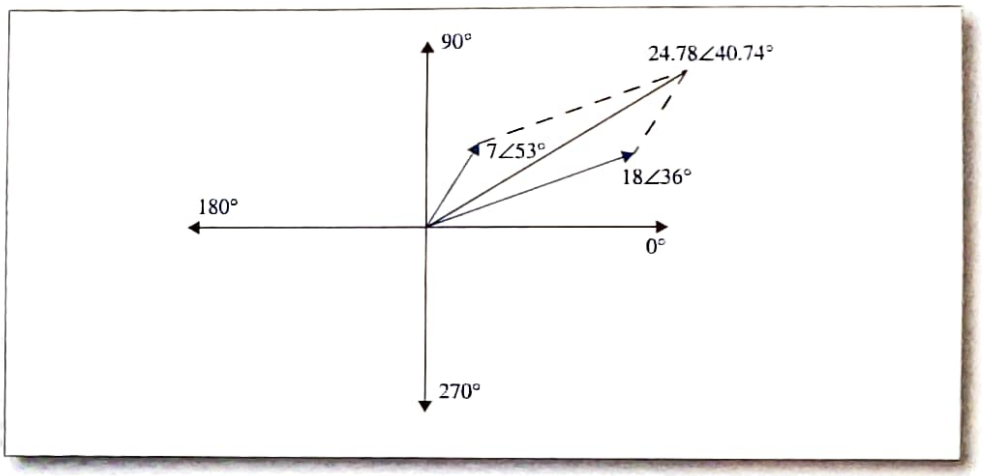
The sum of the two voltages can be converted to the time domain to obtain:

$$v_1 + v_2 = 24.78\sin(10t + 40.74°)\ \text{V}$$

The graphical representation of the given phasors and their sum is shown in the phasor diagram in Figure 4–14.

FIGURE **4–14**

Phasor Diagram for Example 4–8



EXAMPLE 4–9 Phasor Mathematics Using the Engineering Calculator

a. Use the calculator and phasor math to perform the indicated phasor addition or subtraction. Represent the sum or difference in the phasor domain.

$$\mathbf{V} = 16\angle -36°\ \text{V} - 25\angle 48°\ \text{V}$$

✓SOLUTION

Since the phasors are given in polar form, we place the calculator in polar form by selecting MODE...COMPLEX FORMAT... POLAR.



The calculation is keyed as: (16 ∠ 36) − (25 ∠ 48) ENTER to obtain: 28.24 ∠ −97.7

so that: $\mathbf{V} = 28.24\angle -97.7°$

b. Use the calculator to perform the indicated phasor addition and subtraction. Represent the sum in the phasor domain.

$$\mathbf{V} = 18\angle 24° + 8\angle -21° - 14\angle 40°$$

✓SOLUTION

The calculation is keyed as: (18 ∠ 24) + (8 ∠ (−)21) − (14 ∠ 40) ENTER to obtain: $13.95 \angle -19.01$

so that:
$$\mathbf{V} = 13.95 \angle -19.01^\circ$$

c. Given $v_1 = 3.8 \sin(4000t + 25.6^\circ)$ V and $v_2 = 7.3 \sin(4000t - 63.2^\circ)$ V, use the calculator to determine the *product* of the two waveforms in the phasor domain.

✓SOLUTION

$$v_1 = 3.8 \sin(4000t + 25.6^\circ) \text{ V} \rightarrow \mathbf{V}_1 = 3.8 \angle 25.6^\circ \text{ V}$$

$$v_2 = 7.3 \sin(4000t - 63.2^\circ) \text{ V} \rightarrow \mathbf{V}_2 = 7.3 \angle -63.2^\circ \text{ V}$$



The calculation is keyed as: (3.8 ∠ 25.6) ⊠ (7.3 ∠ (−)63.2) ENTER to obtain: $27.74 \angle -37.6$

so that:
$$(\mathbf{V}_1)(\mathbf{V}_2) = 27.74 \angle -37.6^\circ \text{ V}^2$$

Converting the solution to the time domain, we obtain:

$$(v_1)(v_2) = 27.74 \sin(4000t - 37.6^\circ) \text{ V}^2$$

PRACTICE EXERCISES 4–3

The following practice exercise calculations should be performed using phasor math and an appropriate engineering calculator. The calculations should be performed in polar form.

a. Add the given sinusoidal voltage waveforms in the phasor domain. Express the answer in polar form in the phasor domain.

$$v_1 = 6 \sin(3500t + 75^\circ) \text{ V}; \quad v_2 = 11.5 \sin(3500t - 42^\circ) \text{ V}$$

b. Multiply the given sinusoidal current waveforms. Express the answer in polar form in the phasor domain.

$$i_1 = 5.3 \sin(8000t + 53^\circ) \text{ mA}; \quad i_2 = 2.6 \sin(8000t + 15^\circ) \text{ mA}$$

c. Subtract v_2 from v_1 and express the answer in polar form in the phasor domain.

$$v_1 = 6.4 \sin(\omega t + 20^\circ) \text{ V}; \quad v_2 = 7.8 \sin(\omega t - 39^\circ) \text{ V}$$

✓ANSWERS

a. $10.27 \angle -10.65^\circ$ V

b. $13.78 \angle 68^\circ$ mA^2

c. $7.09 \angle 90.38^\circ$ V

Note that answers may vary slightly due to rounding differences.

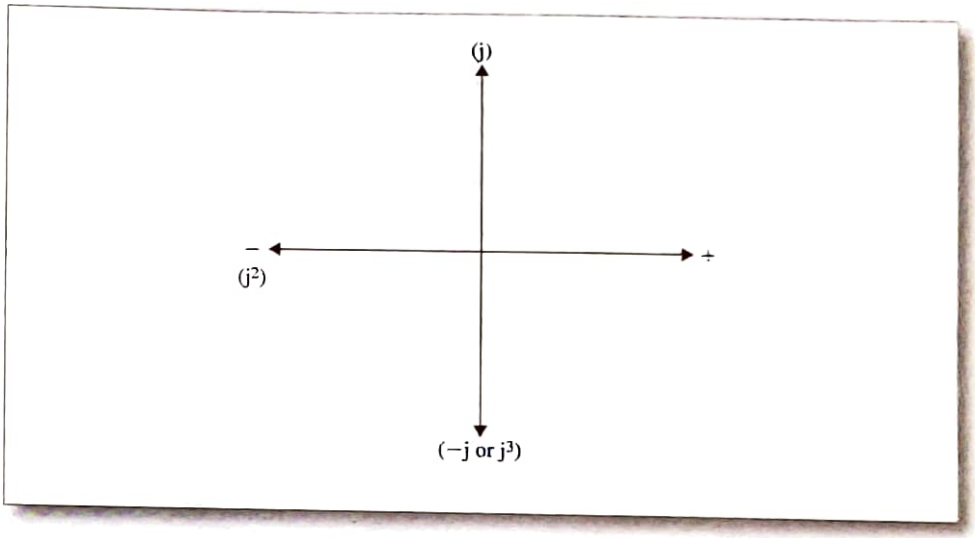
(Complete solutions to the Practice Exercises can be found in Appendix A.)

4.7 THE PHASOR IN RECTANGULAR FORM

An alternate form of phasor representation in the phasor domain often used by engineers expresses a phasor as the sum of its rectangular components. The symbol j is used to indicate that a phasor has been revolved 90° counterclockwise.

The axes of the phasor diagram in rectangular form using the symbol j to indicate 90° of counterclockwise revolution can be labeled as indicated in Figure 4–15.

FIGURE 4–15
The Phasor Domain with Axes Labeled in Rectangular Form

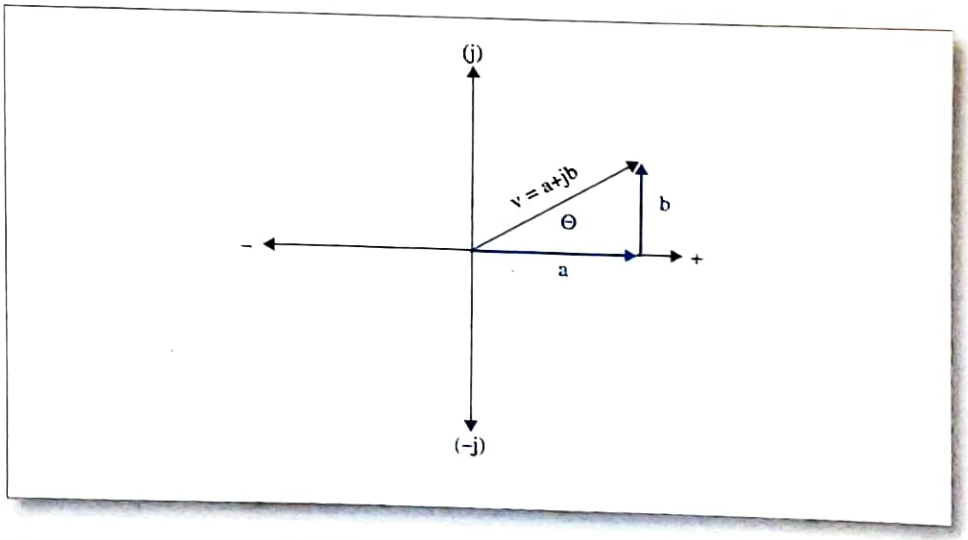


As indicated in Figure 4–15, the symbol j^2 could be used to indicate 180° of revolution; however, the minus sign is the more common notation because j^2 is defined to be equal to -1. Likewise, j^3 could be used to indicate 270°, however, $-j$ is the more common notation.

Consider the phasor **V** in Figure 4–16. Since the rectangular component a is shown in the positive direction along the horizontal axis, it is represented as **a** in rectangular notation. Since the rectangular component b *is* revolved 90° counterclockwise from the horizontal axis, it is represented in rectangular notations as jb. The phasor **V** is the sum of its rectangular components and may therefore be written as:

$$\mathbf{V} = \mathrm{a} + j\mathrm{b} \text{ (rectangular form)}$$

FIGURE 4–16
A Phasor in Rectangular Form



Right-angle trigonometry can be used to determine the peak value or amplitude of the voltage phasor **V** and its phase angle θ respectively as:

$$\mathbf{V} = \sqrt{a^2 + b^2} \tag{4.13}$$

$$\theta = \tan^{-1}(b/a) \tag{4.14}$$

and:

EXAMPLE 4–10 A Phasor in Rectangular Form and Polar Form

A voltage phasor is given in rectangular form as $\mathbf{V} = 6 + j8$. Determine the peak value of the phasor and its phase angle. Express the phasor in polar form. Sketch the phasor in both rectangular form and polar form.

✓SOLUTION

Applying Equations (4.13) and (4.14), the peak value (amplitude) and phase angle may be determined respectively as:

$$\mathbf{V}_P = \sqrt{6^2 + 8^2} = 10 \text{ V}$$

$$\theta = \tan^{-1}(b/a) = \tan^{-1}(8/6) = 53.13°$$

The phasor may be expressed in polar form as:

$$\mathbf{V} = 10 \angle 53.13° \text{ V}$$

The phasor $\mathbf{V} = 6 + j8$ is shown in rectangular form in Figure 4–17a and in polar form as $10 \angle 53.13°$ in Figure 4–17b.

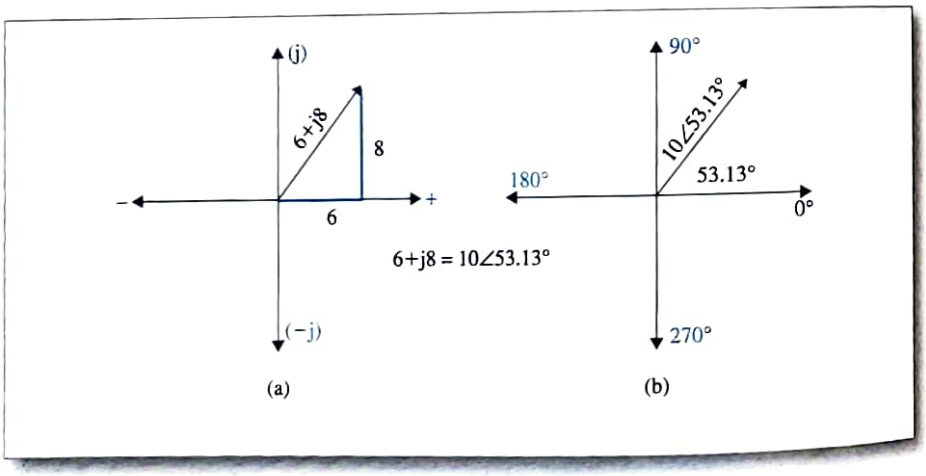


FIGURE 4–17
V = 10 ∠ 53.13° V in Polar Form is Equivalent to V = 6 + *j*8 in Rectangular Form

The phasors in Figure 4–17 a and b are equivalent in every way, but simply expressed in different form.

CALCULATOR

Converting a phasor from rectangular form to polar form can be performed using the TI-89 or similar calculator. The phasor given in Example 4–10 may be converted from the given rectangular form to polar form using the following keystrokes.

Select MODE...COMPLEX FORMAT...POLAR.

Key: (6 + 8 *i*) ENTER to obtain: $10 \angle 53.13°$. Note the symbol *i* is keyed as: 2nd CATALOG.

Converting a phasor from polar form to rectangular form may also be performed as:

Select [MODE]...COMPLEX FORMAT...RECTANGULAR.

Key: ([10] ∠ [53].[13]) [ENTER] to obtain: $6+8i$ (substituting j for i, we obtain $6+j8$).

Note: To avoid confusion, the symbol j is substituted for the calculator symbol i because i is the symbol chosen to represent current.

PRACTICE EXERCISES 4–4

Convert each phasor given in polar form to an equivalent phasor in rectangular form. Convert each phasor given in rectangular form to an equivalent phasor in polar form.

a. $8\angle 58°$ V

b. $36\angle -35°$ mA

c. $14+j24$ V

d. $16-j12.5$ mA

✓ANSWERS

a. $4.236+j6.78$ V

b. $29.49-j20.64$ mA

c. $27.78\angle 39.74°$ V

d. $20.30\angle -37.99°$ mA

(Complete solutions to the Practice Exercises can be found in Appendix A.)

EXAMPLE 4–11 Addition of Phasors in Rectangular Form

Use the calculator and phasor math to add the given phasors in rectangular form. Also determine the sum in polar form.

$$\mathbf{V}_1 = 6+j8 \text{ V}; \quad \mathbf{V}_2 = 3+j5 \text{ V}$$

✓SOLUTION



The TI-89 calculator can perform mathematical computations involving phasors expressed in either rectangular or polar form.

Select [MODE]...COMPLEX FORMAT...RECTANGULAR.

Key: ([6] [+] $8i$) [+] ([3] [+] $5i$) [ENTER] to obtain: $9 + 13i$ (substituting j for i, we obtain $9+j13$).

Note that phasor addition in rectangular form is accomplished simply by adding the *respective parts* of the phasors ($6 + 3 = 9$ and $8i + 5i = 13i$). Before the invention of the engineering calculator, phasors were required to be expressed in rectangular form in order to perform the operations of addition or subtraction.

Although the calculator has minimized the need to express phasors in rectangular form, many engineers continue to use the rectangular form to represent phasor quantities in electronic circuits.

4.8 PLOTTING SINE WAVES WITH MATLAB

It is often necessary to create a mathematical or analytical expression for a sinusoidal waveform that is observed on an oscilloscope, so that required mathematical calculations can be made. It is equally important to create a graphical representation of a sinusoidal waveform if its analytical expression is known.

The modern digital oscilloscope and computer plotting software such as MATLAB have eliminated much of the labor formerly required to create analytical expressions from graphical presentations. Digital oscilloscopes directly display waveform parameters such as frequency, period, peak voltage, and phase angles, while MATLAB can plot functions given appropriate parameters for the waveforms. Many engineering solutions typically combine both mathematical calculations and graphical analysis using software such as MATLAB.

EXAMPLE 4–12 Plotting a Sine Wave with MATLAB

Create a MATLAB plot for the given sinusoidal voltage waveform:

$$v = 3.2 \sin(50t + 45°) \text{ V}$$

✓SOLUTION

The following commands will provide a basic plot of the given sine function. Note the phase angle (45°) is entered into MATLAB as pi/4 rad.

```
EDU ≫ t = linspace(−pi/50,4*pi/50,400);   (scales the horizontal axis)
EDU ≫ v=3.2*sin(50*t+pi/4);               (enters the desired function)
EDU ≫ plot(t,v)                           (plots v as a function of time)
```

The basic MATLAB plot is shown in Figure 4–18.

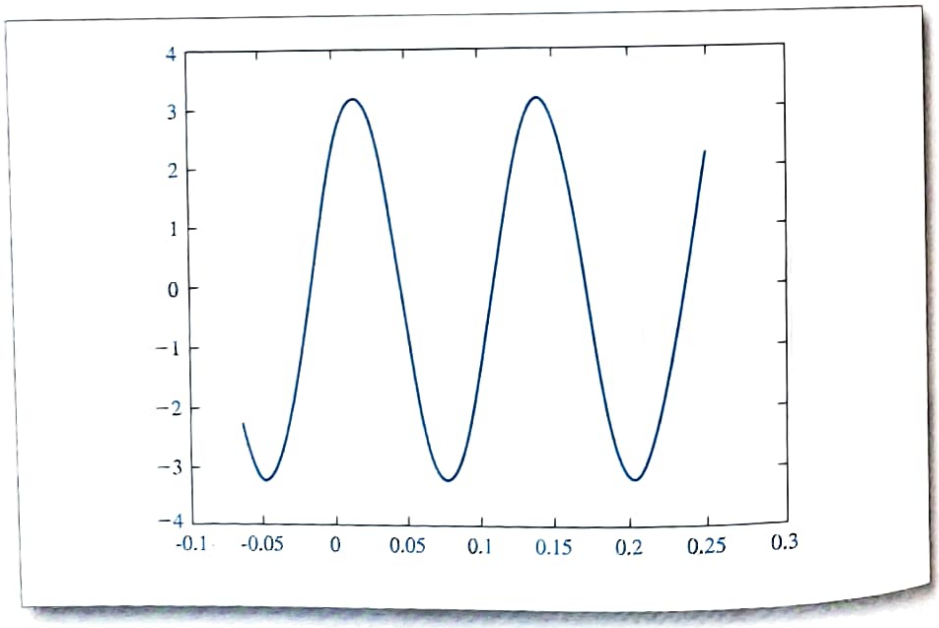


FIGURE 4–18
MATLAB Plot of $v = 3.2 \sin(50t + 45°)$ V

The command t = linspace(−pi/50,4*pi/50,400) scales the horizontal axis to cover the range in time that is equivalent to the angular range of $-\pi$ to 4π rad in 400 steps.

The first two terms of the linspace command are properly scaled by dividing by the radian frequency (ω = 50 rad/s) as indicated in the linspace command.

Additional features such as a title, axes labels, text, arrows, and grid lines can be added to the basic MATLAB plot using the EDIT feature available on the plot figure menu. Several additional enhancements to the basic plot are shown in Figure 4–19.

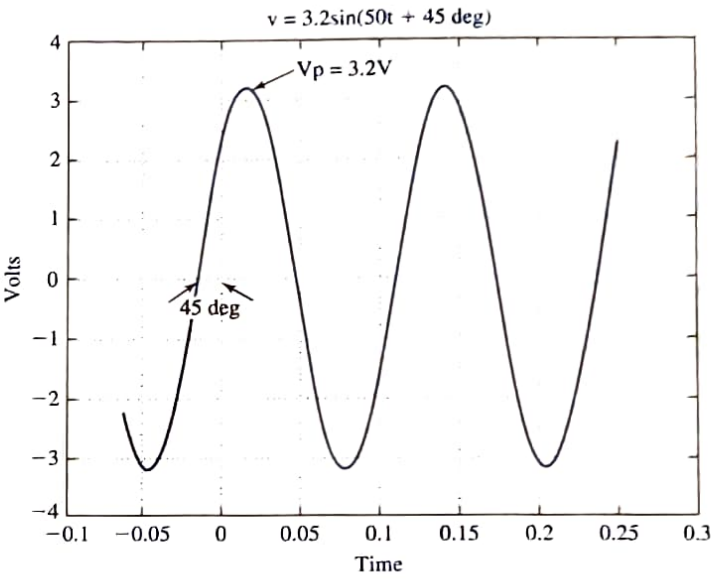


FIGURE 4–19
A MATLAB Plot with Axes Labels, Title, and Text

Note that the plot does not begin a cycle at $t = 0$. The given analytical expression indicates that the waveform is shifted to the left by 45° (π/4 rad) as indicated in Figure 4–19.

4.9 THE RESISTOR IN THE TIME DOMAIN

Assume that the voltage across the resistor in Figure 4–20 is sinusoidal and expressed in the time domain as:

$$v_R = V_p \sin(\omega t + \theta) \tag{4.15}$$

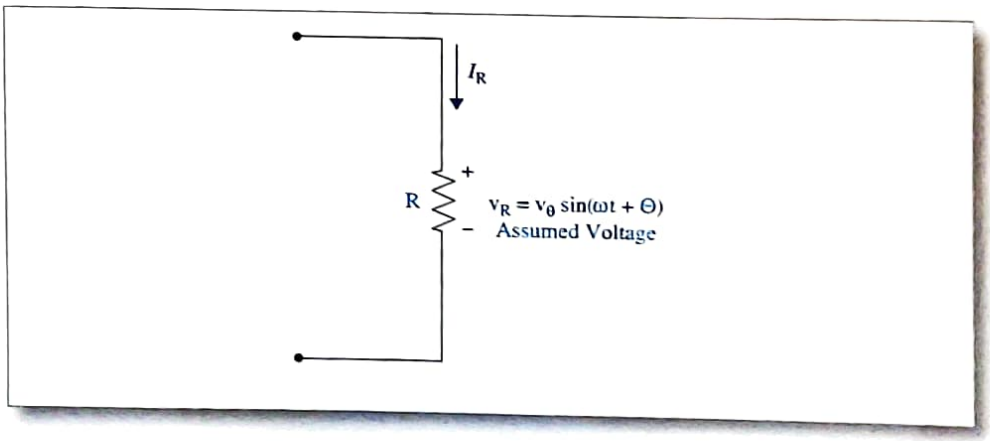


FIGURE 4–20
Circuit Diagram of a Resistor with Assumed Voltage $v_R = V_p \sin(\omega t + \theta)$

Ohm's law can be used to obtain an expression for the current flowing through the resistor as:

$$i_R = \frac{v_R}{R} = \frac{V_P \sin(\omega t + \theta)}{R} = \frac{V_P}{R} \sin(\omega t + \theta) \tag{4.16}$$

The coefficient of the sine function in Equation (4.16) indicates that the peak value of the current flowing through the resistor is:

$$I_P = \frac{V_P}{R}$$

Equation (4.16) also indicates that the current through the resistor has the same radian frequency (ω) and the same phase angle (θ) as the voltage across the resistor.

Waveforms that have the same phase angle have a phase difference of zero and are said to be in phase.

The phase relationship between sinusoidal current and voltage for a resistor is illustrated in Figure 4–21.

MATLAB

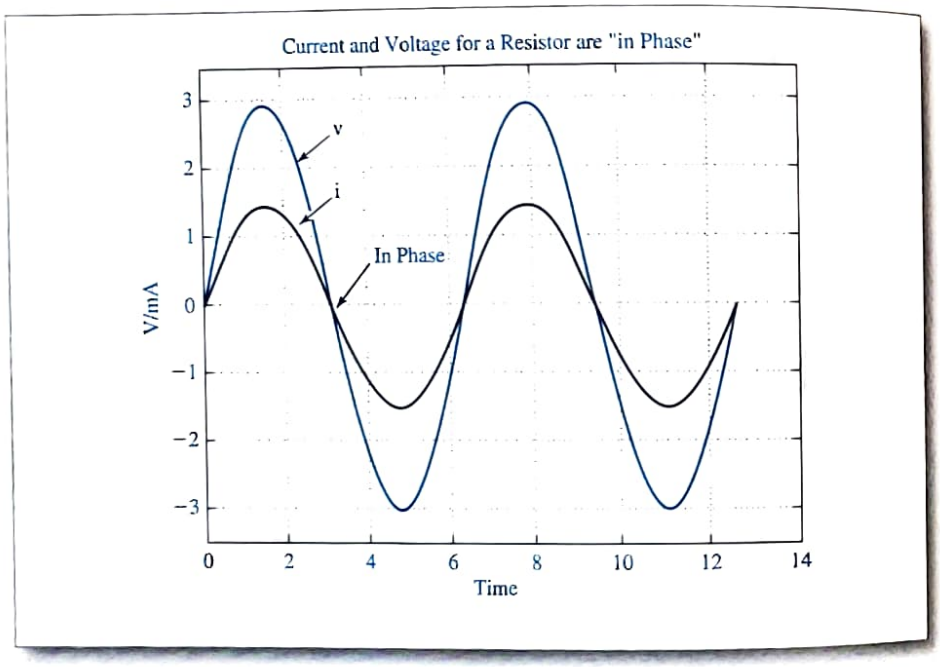


FIGURE 4–21

Resistor Current and Voltage are "In Phase"

EXAMPLE 4–13 A Resistor in the Time Domain

The voltage across a resistor is given as $v_R = 16 \sin(\omega t)$ V, and the resistor value is given as 2 kΩ. Determine the current through the resistor in the time domain.

✓SOLUTION

It is an excellent practice to sketch a schematic drawing that includes important information such as schematic symbols, component connections, voltage polarities, current flow directions, and the values of all circuit components.

The schematic drawing of the problem statement is shown in Figure 4–22.

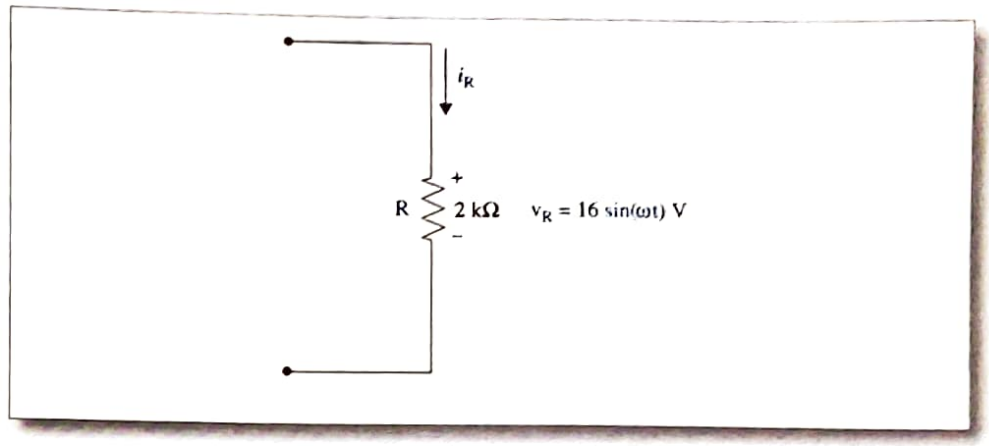


FIGURE 4–22
Circuit Diagram for Example 4–13

Ohm's law can be used to calculate the peak value of current as:

$$I_P = \frac{V_P}{R} = \frac{16 \text{ V}}{2 \text{ k}\Omega} = 8 \text{ mA}$$

The current as a function of time can be expressed as:

$$i_R = 8 \sin(\omega t) \text{ mA}$$

The phase difference between the current through and voltage across a resistor is 0°. Noting that the given voltage in this example has a phase angle of 0°, the current must also have a phase angle of 0°.

A sinusoidal voltage will reverse polarity on every half cycle. As the voltage reverses polarity, the current through the resistor will reverse direction as well. The voltage polarities and current directions indicated on a schematic are references at an instant of time. It is understood that the voltage polarities and current directions will reverse with each successive half cycle.

The voltage and current waveforms are shown in the MATLAB plot in Figure 4–23.

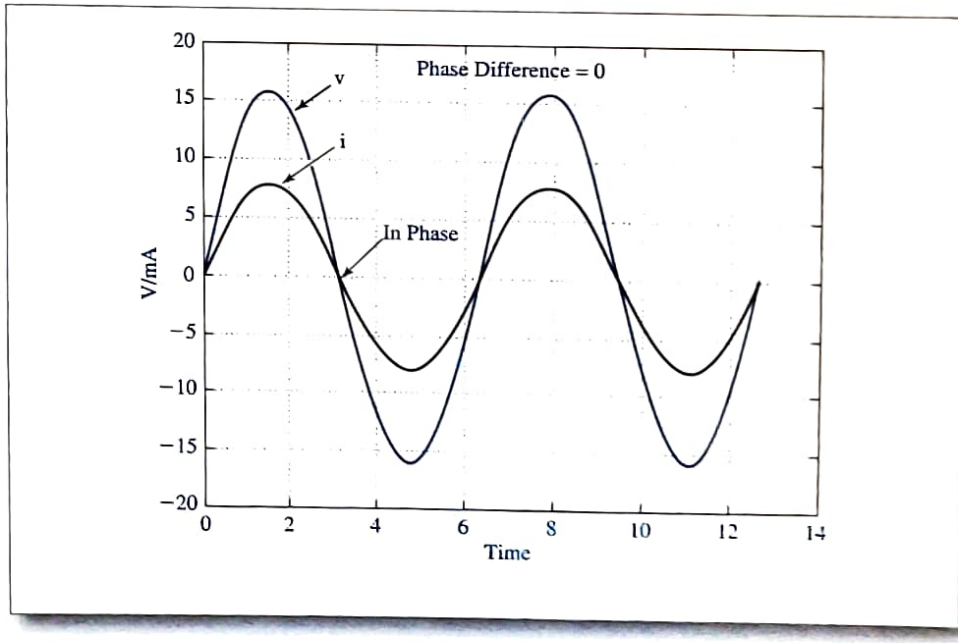


FIGURE 4–23
Voltage and Current Waveforms for Example 4–13

Note that the current i_R has the same radian frequency (ω) and cyclic frequency (f) as the applied voltage in Figure 4–23. Also note that the voltage and current waveforms

cross the horizontal axis at the same time, indicating that the phase difference between the current and voltage is 0.

EXAMPLE 4–14 A Resistor in the Time Domain

The current through a resistor is given as $i = 20 \sin(50t + 45°)$ mA; and the resistance value is given as $R = 3$ kΩ. Determine the voltage across the resistor in the time domain.

✓SOLUTION

The schematic is shown in Figure 4–24.

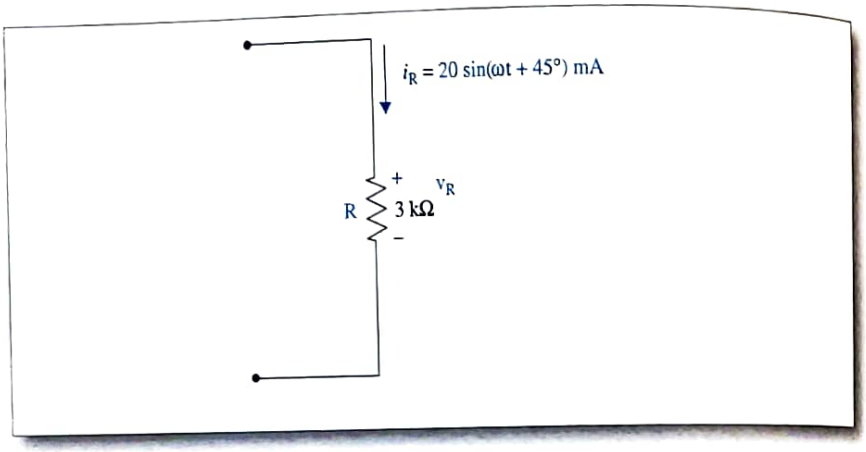


FIGURE 4–24
Circuit Diagram for Example 4–14

Ohm's law can be used to calculate the peak voltage as:

$$V_P = I_P R = (20 \text{ mA})(3 \text{ k}\Omega) = 60 \text{ V}$$

The voltage across the resistor can be expressed as a function of time as:

$$v_R = 60 \sin(50t + 45°) \text{ V}$$

Note that once again the resistor current and voltage have the same phase angle of 45°.

The waveforms are shown in the MATLAB plot in Figure 4–25.

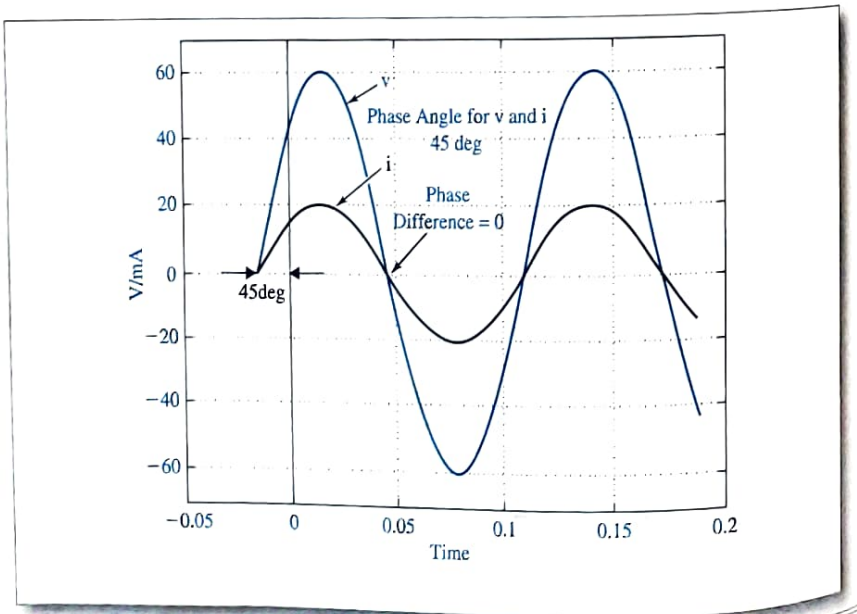


FIGURE 4–25
Voltage and Current Waveforms for Example 4–14

Note that the current and voltage waveforms *do not begin a cycle at* $t = 0$. Both waveforms have been shifted to the left from the origin. The analytical expressions for the current and voltage indicate that the shift is 45° ($\pi/4$ rad). The 45° phase shift is also indicated in Figure 4–24. *The phase angle of the current and voltage is 45° so that the phase difference between the two waveforms is 0°.*

PRACTICE EXERCISES 4–5

a. The current through a 4 kΩ resistor is given as $i_R = 5.3 \sin(300t + 18°)$ mA. Determine the expression for the voltage across the resistor as a function of time.

b. The voltage across a 10 kΩ resistor is given as $v_R = 24 \sin(2000t + 120°)$ V. Determine the expression for the current through the resistor as a function of time.

c. A voltage source is given as $v_S = 15 \sin(300t)$ V. Determine how much resistance is required to limit the source current to $i_S = 20 \sin(300t)$ mA.

✓ANSWERS

a. $21.2 \sin(300t + 18°)$ V

b. $2.4 \sin(2000t + 120°)$ mA

c. 750 Ω

(Complete solutions to the Practice Exercises can be found in Appendix A.)

4.10 THE RESISTOR IN THE PHASOR DOMAIN

Mathematical calculations involving sinusoidal waveforms are more easily performed in the phasor domain than in the time domain. In order to perform mathematical operations in the phasor domain, however, it is necessary to convert all sinusoidal voltages and currents from their time domain expressions to equivalent phasor domain expressions. In addition, a phasor representation for each circuit component must be determined as well. Voltages and currents may be expressed in the phasor domain in either polar or rectangular form. Circuit components also may be represented in either polar or rectangular form.

A resistor is represented in polar form in the phasor domain as:

$$\mathbf{R} = R\angle 0° \tag{4.17}$$

A resistor is represented in rectangular form in the phasor domain as:

$$\mathbf{R} = R + j0 \tag{4.18}$$

Note that the symbol R without bold-face print is the resistance value of the resistor.

Using the phasor notation $\mathbf{R} = R\angle 0°$ to represent a resistor, the proper phase relationship between current and voltage for the resistor is maintained when making calculations in the phasor domain.

EXAMPLE 4–15 The Resistor in the Phasor Domain

In Example 4–13, the voltage across a 2 kΩ resistor is given as $v = 16 \sin(\omega t)$ V. Calculate the current through the resistor using phasor mathematics, and compare the solution to the one obtained in Example 4–13.

✓SOLUTION

Converting the given sinusoidal voltage to polar form in the phasor domain, we obtain:

$$v_R = 16 \sin(\omega t) \text{ V} \rightarrow \mathbf{V} = 16\angle 0° \text{ V}$$

The resistor is represented in polar form in the phasor domain using Equation (4.17) as:

$$\mathbf{R} = 2\angle 0° \text{ k}\Omega$$

Ohm's law can be used to determine the resistor current in polar form in the phasor domain as:

$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{R}} = \frac{16\angle 0° \text{ V}}{2\angle 0° \text{ k}\Omega} = 8\angle 0° \text{ mA}$$

The calculation reveals that the current through the resistor has a magnitude (peak value) of 8 mA and that the phase angle of the current is 0°. Since the phase angle of the current and voltage are both 0°, the phase difference is 0°.

The current expressed in the phasor domain can be converted to the time domain as:

$$i_R = 8 \sin(\omega t) \text{ mA}$$

The expression for i_R is the same result obtained in Example 4–13 where the calculations were performed in the time domain.

EXAMPLE 4–16 The Resistor in the Phasor Domain

Given $i_R = 20 \sin(50t + 45°)$ mA and $R = 3$ kΩ, determine the voltage across the resistor as a function of time.

✓SOLUTION

Converting i_R to polar form in the phasor domain, we obtain:

$$\mathbf{I} = 20\angle 45° \text{ mA}$$

The resistor is represented in polar form in the phasor domain as:

$$\mathbf{R} = 3\angle 0° \text{ k}\Omega$$

Ohm's law can be used to determine the voltage across the resistor as:

$$\mathbf{V} = \mathbf{IR} = (20\angle 45° \text{ mA})(3\angle 0° \text{ k}\Omega) = 60\angle 45° \text{ V}$$

Converting the voltage phasor to the time domain, we obtain:

$$v_R = 60 \sin(50t + 45°) \text{ V}$$

The expression for v_R is the same result obtained in Example 4–14 where the calculations were performed in the time domain.

The phasor diagrams for the current and voltage phasors in Examples 4–15 and 4–16 are shown in Figure 4–26 a and b respectively.

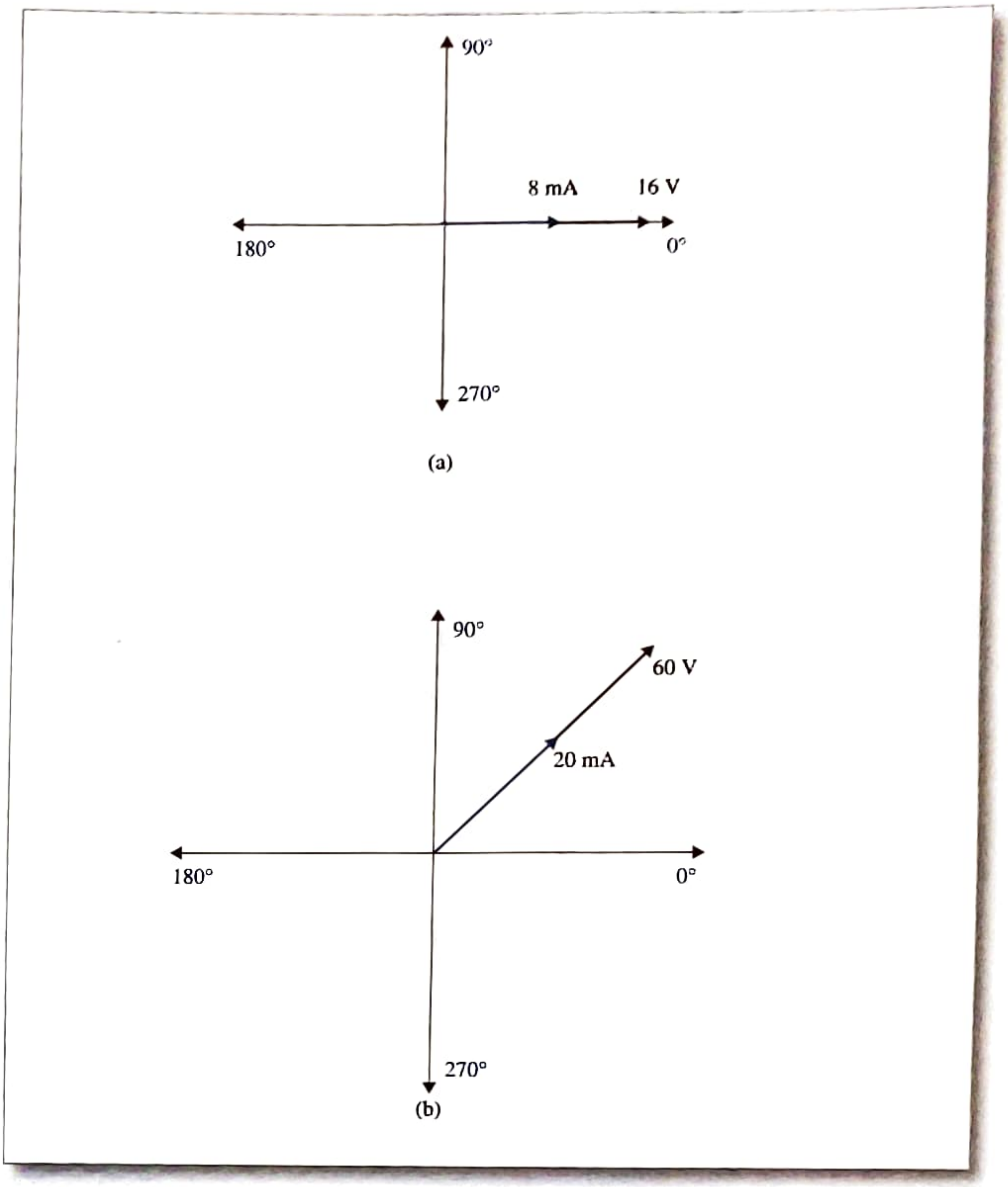


FIGURE **4–26**

Phasor Diagrams for the Current and Voltage Phasors in Example 4–15 and Example 4–16

The phasor diagrams in Figure 4–26 clearly indicate that the voltage across a resistor has the same phase angle as the current through the resistor so that the phase difference is 0.

PRACTICE EXERCISES 4–6

a. The voltage across a 4.7 kΩ resistor is $13 \sin(480t + 29°)$ V. Calculate the current through the resistor in polar form.

b. The current through a 500 Ω resistor is $41 \sin(1000t + 68°)$ mA. Calculate the voltage across the resistor in polar form.

✓ANSWERS

a. $\mathbf{I} = 2.77 \angle 29°$ mA

b. $\mathbf{V} = 20.5 \angle 68°$ V

(Complete solutions to the Practice Exercises can be found in Appendix A.)

The important characteristics of a resistor may be summarized as:

- The resistance of a resistor is independent of frequency.
- The magnitudes of current and voltage for a resistor are related by Ohm's law:

$$I_R = \frac{V_R}{R};\ V_R = I_R R$$

- The phase difference between the voltage and current for a resistor is 0°.
- A resistor is represented in the phasor domain as $\mathbf{R} = R\angle 0°$ (polar form) or $\mathbf{R} = R + j0$ (rectangular form).

4.11 THE INDUCTOR IN THE TIME DOMAIN

An inductor is a circuit component that is constructed by winding turns of copper wire around an iron core, as illustrated in Figure 4–27. As current passes through an inductor, an expanding magnetic field will develop around the device. As the expanding magnetic field cuts across the windings of the inductor, a voltage is induced across the inductor, as illustrated in Figure 4–27.

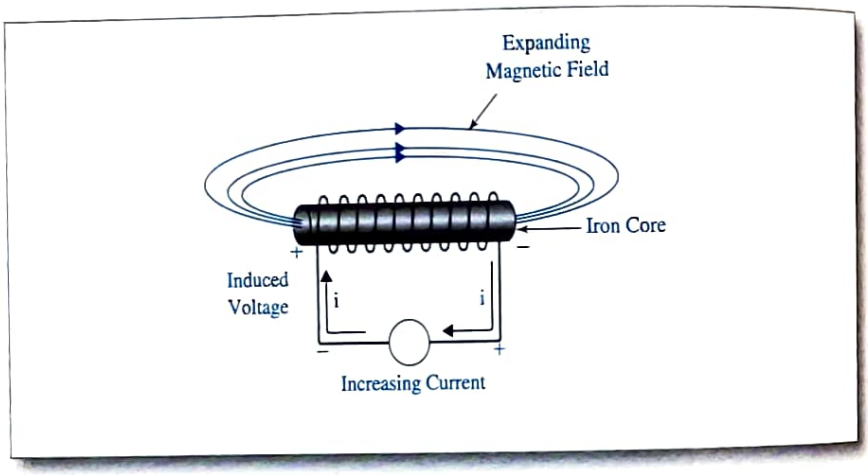


FIGURE 4–27

Increasing Current through an Inductor Produces an Expanding Magnetic Field around the Inductor

As the current diminishes, the magnetic field will begin to collapse, again cutting across the windings of the inductor. The collapsing magnetic field will also induce a voltage across the inductor, but with the opposite polarity. In both cases, *relative motion* exists between the inductor and the magnetic field surrounding the device, resulting in an induced voltage across the inductor.

A change in the current flowing through an inductor will produce either an expanding or collapsing magnetic field that will induce a voltage across the inductor. Since a sinusoidal current is continuously changing, it will induce a continuous voltage across the inductor. The induced voltage will also be sinusoidal and will have the same frequency as the current.

The polarity of the voltage across an inductor is determined by Lenz's law, which states that an induced effect will always oppose the cause that produced it. It follows that the polarity of the induced voltage across an inductor will always oppose a change of current through the device.